



EXPERIMENT 8

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Subject Name: ADBMS

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Experiment 8 -- To study and implement Views

Aim : To study and learn the about the Views, grant or revoke submission.

Question :8 Part A (i): Complex View

SQL Query:

```
CREATE TABLE customer_master
```

```
(  
    customer_id VARCHAR(5) PRIMARY KEY,  
    full_name VARCHAR(50) NOT NULL,  
    city VARCHAR(30)  
);
```

```
CREATE TABLE product_catalog
```

```
(  
    product_id VARCHAR(5) PRIMARY KEY,  
    product_name VARCHAR(50) NOT NULL,  
    unit_price NUMERIC(10,2) NOT NULL  
);
```

```
CREATE TABLE sales_orders
```

```
(
```

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```
order_id SERIAL PRIMARY KEY,  
product_id VARCHAR(5)  
    REFERENCES product_catalog(product_id),  
  
quantity INT NOT NULL,  
  
customer_id VARCHAR(5)  
    REFERENCES customer_master(customer_id),  
  
discount_percent NUMERIC(5,2),  
  
order_date DATE NOT NULL  
);
```

```
INSERT INTO customer_master  
(customer_id, full_name, city)  
VALUES  
('C1', 'Amit Sharma', 'Delhi'),  
('C2', 'Priya Verma', 'Mumbai'),  
('C3', 'Ravi Kumar', 'Bangalore'),  
('C4', 'Neha Singh', 'Kolkata'),  
('C5', 'Arjun Mehta', 'Hyderabad');
```

```
INSERT INTO product_catalog  
(product_id, product_name, unit_price)  
VALUES  
('P1', 'Smartphone X100', 25000),  
('P2', 'Laptop Pro 15', 65000),  
('P3', 'Wireless Earbuds', 5000),
```

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('P4', 'Smartwatch Fit', 30000),
('P5', 'Gaming Console', 45000);

```
INSERT INTO sales_orders
(product_id, quantity, customer_id, discount_percent, order_date)
VALUES
('P1', 2, 'C1', 5, '2025-09-01'),
('P2', 1, 'C2', 10, '2025-09-02'),
('P3', 3, 'C3', 0, '2025-09-03'),
('P4', 1, 'C4', 8, '2025-09-04'),
('P5', 1, 'C5', 12, '2025-09-05'),
('P1', 1, 'C2', 5, '2025-09-06'),
('P3', 2, 'C1', 0, '2025-09-07');
```

```
CREATE OR REPLACE VIEW vw_sales_report as
select t.order_id, cc.full_name, t.quantity, t.final_am as final_amount
FROM (
    select so.discount_percent, so.product_id, so.customer_id, so.order_id, so.quantity, (
        (pc.unit_price * so.quantity * so.discount_percent )/100
    ) as final_am
FROM sales_orders as so
JOIN product_catalog as pc
ON so.product_id = pc.product_id
) t
JOIN customer_master as cc
ON cc.customer_id = t.customer_id
```

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```
SELECT * from vw_sales_report
```

```
CREATE USER Exp_8_user with PASSWORD '1234';  
GRANT SELECT ON vw_sales_report TO Exp_8_user;
```

Question: Part A (ii) Materialized view for Category_report_generation
SQL Query:

```
DROP TABLE IF EXISTS category_master CASCADE;  
DROP TABLE IF EXISTS product_master CASCADE;  
DROP TABLE IF EXISTS customer_master CASCADE;  
DROP TABLE IF EXISTS order_details CASCADE;
```

```
CREATE TABLE category_master (  
    category_id VARCHAR(5) PRIMARY KEY,  
    category_name VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE product_master (  
    product_id VARCHAR(5) PRIMARY KEY,  
    product_name VARCHAR(50) NOT NULL,  
    category_id VARCHAR(5)  
    REFERENCES category_master(category_id)  
);
```

```
CREATE TABLE customer_master (  
    customer_id VARCHAR(5) PRIMARY KEY,  
    customer_name VARCHAR(50) NOT NULL,  
    category_id VARCHAR(5)  
    REFERENCES category_master(category_id)  
);
```

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```
customer_id VARCHAR(5) PRIMARY KEY,  
customer_name VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE order_details (  
    order_id VARCHAR(5) NOT NULL,  
    customer_id VARCHAR(5)  
        REFERENCES customer_master(customer_id),  
    product_id VARCHAR(5)  
        REFERENCES product_master(product_id),  
    quantity INT NOT NULL,  
    unit_price NUMERIC(10,2) NOT NULL  
);
```

```
INSERT INTO category_master  
(category_id, category_name)  
VALUES  
( 'C1', 'Electronics'),  
( 'C2', 'Clothing'),  
( 'C3', 'Books');
```

```
INSERT INTO product_master  
(product_id, product_name, category_id)  
VALUES  
( 'P101', 'Laptop', 'C1'),  
( 'P102', 'Headphones', 'C1'),  
( 'P201', 'Jacket', 'C2'),
```

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```
('P202', 'Shoes', 'C2'),  
( 'P301', 'SQL Book', 'C3');
```

```
INSERT INTO customer_master
```

```
(customer_id, customer_name)
```

```
VALUES
```

```
('U1', 'Aman'),
```

```
('U2', 'Priya'),
```

```
('U3', 'Rahul'),
```

```
('U4', 'Simran');
```

```
INSERT INTO order_details
```

```
(order_id, customer_id, product_id, quantity, unit_price)
```

```
VALUES
```

```
('O101', 'U1', 'P101', 1, 5000),
```

```
('O102', 'U2', 'P102', 2, 1000),
```

```
('O103', 'U1', 'P201', 1, 3000),
```

```
('O104', 'U3', 'P202', 2, 2000),
```

```
('O105', 'U4', 'P301', 3, 500),
```

```
('O106', 'U2', 'P101', 1, 5000),
```

```
('O107', 'U3', 'P201', 1, 3000),
```

```
('O108', 'U4', 'P102', 1, 1000);
```

```
DROP MATERIALIZED VIEW mw_category_report
```

```
EXPLAIN ANALYZE
```

```
CREATE MATERIALIZED VIEW mw_category_report
```

```
AS
```

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```
select cm.category_id as CatID,  
       cm.category_name as CatName,  
       SUM(od.quantity) as TOTAL_PRODUCT,  
       COUNT(DISTINCT od.order_id) as TOTAL_ORDERS,  
       COUNT(DISTINCT od.customer_id) as Unique_Customer,  
       SUM(od.quantity * od.unit_price) as TOTAL_REVENUE,  
       ROUND( SUM(od.quantity * od.unit_price)/COUNT(DISTINCT od.order_id) ,2) as AVG_REVENUE,  
       (CASE WHEN SUM(od.quantity * od.unit_price) >= 10000 THEN 'Excellent'  
            WHEN SUM(od.quantity * od.unit_price) >= 6000 THEN 'Good'  
            WHEN SUM(od.quantity * od.unit_price) >= 3000 THEN 'Average'  
            ELSE 'Low' END) as Category_performance  
from Order_details as od  
JOIN product_master as pm  
ON od.product_id = pm.product_id  
JOIN category_master as cm  
ON cm.category_id = pm.category_id  
GROUP BY cm.category_id, cm.category_name  
  
SELECT * FROM mw_category_report
```

Learning Outcome

After completing this experiment, the learner will be able to:

1. **Understand and implement Views**
2. **Apply aggregate functions** such as COUNT() and DISTINCT to get unique customer;
3. **Use joins and filtering techniques** to combine data from multiple tables and generate meaningful reports.